

Installed Ground Truth for Training-Data Attribution: Filtering on Gradient-Based Attribution Scores Removes None of the Effect

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Abstract

Two corpora with an identical premise specification but different surface forms installed sharply divergent belief effects: +0.0072 for approximately 740-word articles versus +0.1574 for approximately 105-word answers, defining the minimum attribution challenge. The long-form estimate is a two-epoch reading with matched off-topic control retrained per seed; only surface form differs in the short-answer corpus. We evaluate gradient-based attribution, without generalising to retrieval or embedding methods, by removing top-ranked documents and retraining. The attributable fraction is the share of a per-source causal effect installed and measured in the corpus that disappears after dose-matched removal, with controls retrained per seed.

The headline result is failure to outperform a model-free length baseline: in every declared budget-by-seed cell, TracIn, TracIn-cosine, and delta-predictability each had an attributable-fraction interval overlapping the corresponding word-count interval under the pre-registered kill condition. This interval-overlap criterion is not a pairwise significance test, and no pairwise contrast was computed. Removability was established by the positive-control ceiling: at the 20% budget, oracle removal by separately measured per-source effect removed roughly half the installed effect, with AF +0.5172 at seed 42 and +0.6208 at seed 7, making null readings for the evaluated methods interpretable. At the 10% budget, TracIn was counterproductive in both seeds—AF -0.9130 and -0.4238—with both intervals wholly negative, so removing top-ranked documents increased the effect.

1 Introduction

Attribution of training-induced belief change is difficult even when source content is held constant. Corpora sharing an identical premise specification but differing only in surface form installed sharply divergent causal potency when expressed as long-form articles or short answers. The long-form estimate is a two-epoch reading with matched off-topic control retrained per seed, and recovering this divergence is the minimum attribution challenge.

This challenge spans a broader ladder of source directness: a conclusion-stating corpus at matched dose installed a moderate effect, while an explicit-stance positive-control contrast provided a higher ceiling despite using a lower token dose than the evidence corpus. The explicit-stance point estimate is derived from recorded arm scores without an inferred interval and represents a positive-control contrast rather than general belief transfer.

The evaluation therefore asks whether gradient-based attribution can identify documents whose removal eliminates the installed effect: top-ranked documents are removed and the model is retrained. Its scope is limited to gradient-based attribution and does not generalise to retrieval or embedding methods.

2 Methods

The exploratory factory_farming study used qwen3-4b with seed 42. The primary attributable-fraction run, attrib_mix_v4, evaluated attrib_mix_v4 checkpoint-23 and m0_multiform checkpoint-24 for transfer and efficacy. The attributable fraction was the share of a per-source causal effect installed and measured in the corpus that disappeared after dose-matched removal of top-ranked documents and retraining, with controls retrained per seed. Blank run-id fields were not asserted.

The short-answer belief run, `ms_arms`, used the same experiment, model, and seed and evaluated `ms_arms` checkpoint-24, `m0_multiform` checkpoint-24, and `explicit_stance_v3_arms` checkpoint-24 for transfer and efficacy. The long-form endpoint run, `matrix_v1_step24`, likewise used the same experiment, model, and seed and evaluated `multiformat_v2_valsplit_fixedq_d93` checkpoint-24, `m0_multiform` checkpoint-24, and `explicit_stance_v3_arms` checkpoint-24 for absorption, transfer, and efficacy. Blank run-id fields were not asserted.

Two-seed replication applied only to declared seed-paired contrast families; single-seed and instrument-specific readings were identified separately. Belief and action contrasts used matched off-topic control netting, without a claim of causal mediation, and raw belief/action checkpoint figures were treated as diagnostic rather than as netted transfer estimates. Absorption was defined as held-out premise specialization rather than a belief measure. Fixed tables, rather than prose, were authoritative for numerical estimates and confidence intervals.

3 Results

Under full fine-tuning, the pre-removal pool effects that formed the attributable-fraction denominators were +0.0167 at seed 42 and +0.0247 at seed 7, with overlapping intervals and no LoRA-style halving. In the cleaner full-fine-tuning off-topic control, the single-seed machinery estimate was -0.0004 at seed 42, and its interval straddled zero.

The positive-control ceiling established removability: at the 20% removal budget, oracle removal by separately measured per-source effect removed roughly half the installed effect, with AF +0.5172 at seed 42 and +0.6208 at seed 7, making null readings for the evaluated methods interpretable. At the 10% budget, oracle AF was -0.1048 at seed 42 and +0.3005 at seed 7, and both intervals straddled zero. Although these point estimates were below the 20% cells, the wide intervals did not establish an ordering or a sublinear response.

At the 10% budget, `TracIn` was counterproductive in both seeds: AF -0.9130 at seed 42 and -0.4238 at seed 7, with both intervals excluding zero on the negative side, indicating that removing top-ranked documents increased the effect. `TracIn-cosine`, the prescribed length-normalisation fix, yielded AF -0.9503 at seed 42 with a wholly negative interval, but AF -0.2043 at seed 7 with an interval straddling zero; thus, its counterproductive reading was single-seed.

At the 20% budget, `TracIn` AF was -0.4687 at seed 42 and +0.1689 at seed 7, while `TracIn-cosine` AF was -0.0528 at seed 42 and +0.2168 at seed 7; all corresponding intervals straddled zero.

At the 10% budget, delta-predictability produced an unresolved seed contradiction rather than an average or a working repair: AF -0.5767 at seed 42 with a wholly negative interval versus +0.3424 at seed 7 with a wholly positive interval. At the 20% budget, AF was +0.1284 at seed 42 and +0.2396 at seed 7, with both intervals straddling zero, so the wider budget did not resolve the 10% seed contradiction.

The model-free word-count baseline yielded AF -0.4891 and -0.1395 at the 10% budget for seeds 42 and 7, respectively, and -0.3019 and +0.1163 at the 20% budget; all four intervals straddled zero. In every declared budget-by-seed cell, the AF intervals for `TracIn`, `TracIn-cosine`, and delta-predictability each overlapped the corresponding word-count baseline interval. This pre-registered kill condition was based on interval overlap, not a pairwise significance test, and no pairwise contrast was computed.

Table 1: Belief-transfer summary for the short-answer `ms_arms` run, whose corpus uses the same premise specification as the long-form corpus but a different surface form. Estimates use matched off-topic control netting for the recorded run and checkpoints. Source: `ms_arms/belief_summary.yaml`.

quantity	recorded value
<code>delta_raw</code>	+0.1548 [+0.1221, +0.1881]
<code>machinery</code>	-0.0026 [-0.0063, +0.0006]
<code>delta_net</code>	+0.1574 [+0.1251, +0.1905]
<code>sensitivity</code>	+0.6521 [+0.5578, +0.7413]
<code>T (delta_net)</code>	+0.2414

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

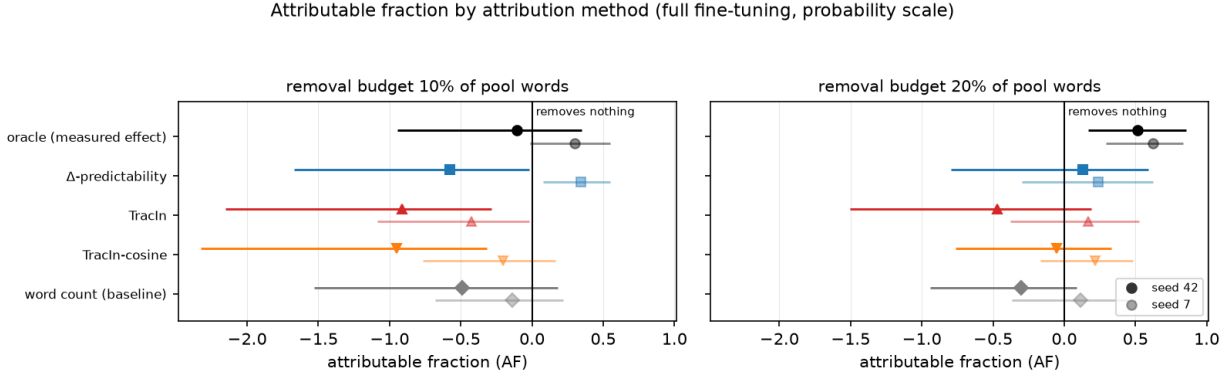


Figure 1: Attributable fraction by removal method and seed under full fine-tuning, with separate panels for the 10% and 20% dose-matched removal budgets and 95% intervals. The zero line denotes removal of none of the installed effect.

4 Discussion

The positive-control ceiling shows that the installed effect was removable: removal by separately measured per-source effect eliminated roughly half of it. This makes null or adverse readings for the evaluated methods interpretable rather than attributable to an intrinsically non-removable endpoint.

Against that ceiling, TracIn was counterproductive at the narrower budget in both seeds, because removing its top-ranked documents increased rather than reduced the effect. The prescribed length-normalisation fix, TracIn-cosine, did not provide a stable remedy: its counterproductive reading appeared in only one seed, while the other seed remained compatible with no effect.

Delta-predictability likewise cannot be treated as a working repair. The two seeds excluded zero in opposite directions, leaving an unresolved contradiction rather than an effect that can be meaningfully averaged.

More broadly, none of the evaluated attribution scores separated from the model-free word-count baseline under the pre-registered kill condition: in every declared budget-by-seed cell, each method’s interval overlapped the corresponding baseline interval. This criterion concerns interval overlap only; it is not a pairwise significance test, and no pairwise contrast was computed.

The resulting verdict is therefore bounded to the attribution scores evaluated for removal, not to content keying generally. No attributable fraction was computed for the retriever, and ranking fidelity is not removal efficacy; delta-predictability’s seed contradiction further shows that a good ordering need not yield a working filter. The evidence also supports no claim about how any evaluated method scores the two content-matched sources.

5 Limitations

The findings are exploratory, and the fixed tables remain authoritative for estimates and confidence intervals. Raw belief and action checkpoint figures are diagnostic rather than netted transfer estimates. Replication across two seeds applies only to the declared seed-paired contrast families; other readings are single-seed or instrument-specific.

The retriever result concerns rank correlations across six sources, not attributable fractions. Although intfloat/e5-base-v2, using chunked inputs and the maximum chunk score, ranked sources in close agreement with installed effects at both polarities and outperformed the model-free length baseline, this does not establish removal efficacy. The mean-over-chunks variant gave the same strong correlation at both polarities, the retriever score was negatively rather than positively associated with word count, and the retriever never saw the trained model.

Accordingly, the conclusions are limited to the attribution scores evaluated for removal rather than content keying generally. Delta-predictability’s opposite-direction seed results illustrate why a good ordering

need not imply a working filter, and the evidence does not establish how any evaluated method scores the two content-matched sources.

6 Conclusion

The gradient-based attribution evaluation asked whether removing top-ranked documents would remove the installed effect. Removal by separately measured per-source effect eliminated roughly half the installed effect, establishing a positive-control ceiling and making null readings for the evaluated methods interpretable.

Nevertheless, TracIn, TracIn-cosine, and delta-predictability failed the pre-registered interval-overlap criterion in every declared budget-by-seed cell because their intervals overlapped the corresponding word-count baseline intervals. This criterion was not a pairwise significance test, and no pairwise contrast was computed. Delta-predictability also remained an unresolved contradiction rather than a working repair: the two seeds excluded zero in opposite directions, so their results cannot be treated as an average.

The verdict is therefore limited to the attribution scores evaluated for removal, not content keying generally. No attributable fraction was computed for the retriever, and ranking fidelity is not removal efficacy; the delta-predictability contradiction shows that a good ordering need not yield a working filter. The evidence supports no claim about how any evaluated method scores the two content-matched sources.

7 Appendix

The attribution challenge begins before removal: corpora with an identical premise specification produced sharply divergent causal potency when expressed as approximately 740-word articles rather than approximately 105-word answers. The long-form estimate is a two-epoch reading with a matched off-topic control retrained per seed, while the short-answer corpus differs only in surface form. Descriptive conclusions at matched dose installed a moderate effect, and the explicit-stance contrast supplied a higher positive-control ceiling despite a lower token dose; that point estimate has no inferred interval and represents explicit stance rather than general belief transfer.

A purpose-trained embedding retriever nevertheless recovered the source ordering closely at both polarities and more faithfully than the model-free length baseline. These are rank correlations over six sources, not attributable fractions. The retriever was intfloat/e5-base-v2, applied to chunked inputs with the maximum chunk score; averaging over chunks also produced strong agreement at both polarities. Its scores were negatively associated with word count, so the ordering was not length-confounded, and the retriever never saw the trained model.

A Supplementary diagnostics

Table 2: Complete ladder of every declared contrast, including installed effects, controls, attributable fractions, and six-source rank correlations with their recorded intervals and qualifications. Attributable fractions and rank correlations are distinct quantities, and the explicit-stance row is an unintervalled positive-control contrast. Source: paper/synthesis.json. *Note.* Rows without intervals are deterministic point contrasts over recorded arm scores.

intervention	net effect	qualification
Installed effect: premises as ~740-word articles (mev)	+0.0072 [+0.0016 , +0.0136]	Two-epoch reading, matched off-topic control retrained per seed.
Installed effect: the SAME premises as ~105-word answers (ms)	+0.1574 [+0.1251 , +0.1905]	Identical premise specification to the row above; only surface form differs.
Installed effect: descriptive conclusions (md)	+0.1106 [+0.0873 , +0.1352]	Conclusion-stating corpus at matched dose.
Installed effect: explicit stance (me)	+0.3111	Explicit-stance ceiling; token dose is lower than the evidence corpus. Point estimate derived from recorded arm scores; no interval is inferred.

intervention	net effect	qualification
Rank correlation with installed effect: embedding retriever, positive polarity	+0.8857 [+0.8857 , +0.9429]	Spearman over 6 sources, not an attributable fraction; no removal was run for this retriever. intfloat/e5-base-v2, chunked, score = max over chunks; the mean-over-chunks variant gives +0.9429 at both polarities. Not length-confounded: rho(score, words) is -0.18 to -0.29.
Rank correlation with installed effect: embedding retriever, negative polarity	+0.9429 [+0.8857 , +0.9429]	Same reading at the other polarity. A rank correlation over 6 sources; ranking fidelity is not removal efficacy.
Rank correlation with installed effect: length baseline, positive polarity	+0.2571 [+0.1429 , +0.4286]	The model-free baseline the retriever is read against, from the same bootstrap. Rank correlation only.
Rank correlation with installed effect: length baseline, negative polarity	+0.2571 [+0.1429 , +0.6571]	Baseline at the other polarity. Rank correlation only.
Pool effect before removal, seed 42 (dB_before)	+0.0167 [+0.0084 , +0.0256]	Denominator of every AF at this seed; full fine-tuning.
Pool effect before removal, seed 7 (dB_before)	+0.0247 [+0.0137 , +0.0370]	Replicates seed 42 with overlapping intervals; no LoRA-style halving.
Off-topic control machinery, seed 42	-0.0004 [-0.0034, +0.0029]	Straddles zero; full-FT control is far cleaner than the LoRA control it replaces.
AF, oracle removal at 20% budget, seed 42	+0.5172 [+0.1686 , +0.8560]	Removal by separately measured per-source effect; the ceiling.
AF, oracle removal at 20% budget, seed 7	+0.6208 [+0.2903 , +0.8355]	Seed-seven replication of the ceiling; spread 0.10.
AF, oracle removal at 10% budget, seed 42	-0.1048 [-0.9458, +0.3511]	Straddles zero. The point estimate is below the 20% budget's, but these wide intervals do not establish an ordering; report as a point-estimate observation only, never as a demonstrated sublinear response.
AF, TracIn removal at 10% budget, seed 42	-0.9130 [-2.1529 , -0.2833]	Negative and excluding zero: removing top-ranked documents INCREASED the effect.
AF, TracIn removal at 10% budget, seed 7	-0.4238 [-1.0877 , -0.0220]	Seed-seven replication of the counterproductive direction.
AF, TracIn removal at 20% budget, seed 42	-0.4687 [-1.5059, +0.1902]	Straddles zero at the wider budget.
AF, TracIn-cosine removal at 10% budget, seed 42	-0.9503 [-2.3249 , -0.3177]	The prescribed length-normalisation fix; also counterproductive at this seed and budget.
AF, TracIn-cosine removal at 20% budget, seed 42	-0.0528 [-0.7635, +0.3306]	Straddles zero at the wider budget.
AF, word-count baseline at 10% budget, seed 42	-0.4891 [-1.5297, +0.1788]	The model-free baseline. Non-separation is asserted as interval OVERLAP with each content-keyed method in the same budget and seed cell; no pairwise contrast was computed and none may be implied.
AF, word-count baseline at 20% budget, seed 42	-0.3019 [-0.9397, +0.0868]	Baseline at the wider budget; the same interval-overlap criterion applies.
AF, word-count baseline at 10% budget, seed 7	-0.1395 [-0.6775, +0.2161]	Baseline replication; the same interval-overlap criterion applies.
AF, delta-predictability at 10% budget, seed 42	-0.5767 [-1.6709 , -0.0183]	Excludes zero NEGATIVE; contradicts the same cell at seed 7. Not a working repair.
AF, delta-predictability at 10% budget, seed 7	+0.3424 [+0.0781 , +0.5479]	Excludes zero POSITIVE; contradicts seed 42. The pair is a seed contradiction, not an average.
AF, delta-predictability at 20% budget, seed 42	+0.1284 [-0.7962, +0.5912]	Straddles zero at the wider budget.
AF, oracle removal at 10% budget, seed 7	+0.3005 [-0.0162, +0.5484]	Seed-seven companion to the sublinear-response row.
AF, TracIn removal at 20% budget, seed 7	+0.1689 [-0.3805, +0.5248]	Straddles zero; completes the TracIn cell block.
AF, TracIn-cosine at 10% budget, seed 7	-0.2043 [-0.7652, +0.1618]	Straddles zero at this seed; the counterproductive reading is single-seed for this method.
AF, TracIn-cosine at 20% budget, seed 7	+0.2168 [-0.1678, +0.4817]	Straddles zero; completes the TracIn-cosine cell block.
AF, word-count baseline at 20% budget, seed 7	+0.1163 [-0.3663, +0.4451]	Completes the baseline block. Every content-keyed cell is compared against these by interval overlap only.
AF, delta-predictability at 20% budget, seed 7	+0.2396 [-0.2996, +0.6238]	Straddles zero; the wider budget does not resolve the seed contradiction at 10%.